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Dedication

With profound appreciation and much love, I would like to thank every person who has contributed to my journey, every soul who has lifted me up, and every heart that has believed in me.

To my mother **Fathia** and my father **Sadok** your unconditional love, sacrifices, support and encouragement Thank you, for being the embodiment of love and strength, I am forever grateful for your love.

To my sister **Feriel**, brothers **Nader, Oday, Roudayna**, our bond is unbreakable, and my dedication to you is everlasting. Together, we are a force of love, support, and lifelong companionship.

To my friends, **Ayoub , Aziz, Iyed , Rathwane, Sami** and all my friends, The memories we've created and the bonds we've formed will forever hold a special place in my heart. Thank you for the endless support, the late-night study sessions, and the shared moments of joy.

With all my love and gratitude,

-Kossay

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I would like also to express my dearest thank to **Hendrik Thureau** and his supportqmlskdfjmqklsdfj

I would like to express our heartfelt appreciation to the entire academic staff of the **Higher Institute of Computer Science and Mathematics Monastir** in . Their significant contributions have played a vital role in our comprehensive education.

Also we would like to sincerely thank all the members of the jury, Mr.**1** and Mrs.**2** , as well as our supervisor, for the honor they bestowed upon us by accepting to evaluate this work.

Finally, I would like to express my gratitude to all those who have helped me, directly or indirectly, in the completion of this project.

Abstract

This report presents the culmination of my final-year project in Information and Communication Technology speciality in Networks and Internet of Things. The project aimed to design and implement a boat monitoring and security system designed to enhance the safety and management of boats, particularly in remote and offshore environments. It provides real-time tracking, environmental monitoring, and intrusion detection, ensuring the boat's condition is continuously supervised. The system utilizes wireless communication to send alerts and location updates to the owner, enabling timely responses to potential threats such as unauthorized access, gas leaks, or abnormal environmental conditions. Powered by a solar-charged battery system, it operates autonomously and reliably, making it a practical solution for boat owners seeking peace of mind and improved security while at sea.

Keywords : IoT, GPS Tracking, ESP32, Boat Monitoring, Remote Alerts, Solar-Powered, Maritime Safety, Mobile Application, Control.

Résumé

Ce mémoire présente l'aboutissement de mon projet de fin d'année en Technologies de l'Information et de la Communication spécialité Réseaux et Internet des Objets. Le projet visait à concevoir et à mettre en œuvre un système de surveillance et de sécurité des bateaux conçu pour améliorer la sécurité et la gestion des bateaux, en particulier dans les environnements éloignés et offshore. Il assure un suivi en temps réel, une surveillance environnementale et une détection des intrusions, garantissant ainsi une surveillance continue de l'état du bateau. Le système utilise la communication sans fil pour envoyer des alertes et des mises à jour de localisation au propriétaire, permettant ainsi de répondre rapidement aux menaces potentielles telles qu'un accès non autorisé, des fuites de gaz ou des conditions environnementales anormales. Alimenté par un système de batterie solaire, il fonctionne de manière autonome et fiable, ce qui en fait une solution pratique pour les propriétaires de bateaux en quête de tranquillité d'esprit et d'une sécurité accrue en mer.

Mots clés : IoT, Suivi GPS, ESP32, Surveillance des Bateaux, Alertes à distance, Alimenté par l'énergie solaire, Sécurité maritime, Application Mobile, Contrôle.

List of abbreviations and acronyms

GPRS	<i>General Packet Radio Service</i>
SIM	<i>Subscriber Identity Module</i>
IOT	<i>Internet of Things</i>
APN	<i>Access Point Name</i>
IDE	<i>Integrated Development Environment</i>
VCC	<i>Voltage Common Collector (or Power Supply Voltage)</i>
GND	<i>Ground</i>
NO	<i>Normally Open</i>
LDR	<i>Light Dependent Resistor</i>
PCB	<i>Printed Circuit Board</i>
DC	<i>Direct Current</i>
AC	<i>Alternating Current</i>
AC	<i>Alternating Current</i>
LoRa	<i>Long Range</i>
UART	<i>Universal Asynchronous Receiver-Transmitter</i>
RXD	<i>Receive Data</i>
TXD	<i>Transmit Data</i>
CAD	<i>Computer-Aided Design</i>

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“Graduation is not the end, it’s the beginning.”
—Senator Orrin Hatch

May 19, 2025



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General Introduction

In recent years, maritime activities have grown significantly, ranging from fishing and transportation to recreational boating. However, with this growth, the need for ensuring the safety, security, and operational efficiency of boats has become increasingly critical. Boats are often exposed to harsh environmental conditions, theft, accidents, and various operational challenges. These issues highlight the necessity for a robust and reliable boat monitoring system that can provide real-time data and alerts to boat owners, enabling them to respond promptly to emergencies and ensure the safety of their vessels.

This project aims to design and develop an autonomous boat monitoring system that utilizes modern technologies such as the Internet of Things (IoT), GPS tracking, and wireless communication. The system will integrate sensors for environmental and security monitoring, including temperature and humidity measurement, gas detection, and motion detection. It will also feature real-time location tracking and power monitoring to ensure the boat's battery status is always known. The system will operate independently using a solar-powered energy supply, ensuring continuous functionality even in remote marine environments where access to electricity is limited.

The primary objective of this project is to enhance the safety and security of small boats by providing boat owners with a cost-effective, real-time monitoring solution. This system will allow owners to track their boat's location, receive alerts about unauthorized access or hazardous gas leaks, and monitor the onboard environmental conditions. Such a system is particularly beneficial for boat owners who leave their vessels unattended for long periods or operate in isolated areas.

The motivation behind this project stems from the growing number of incidents involving boat theft, accidents at sea, and environmental damage due to undetected gas leaks or battery failures. These concerns underscore the need for a comprehensive monitoring system capable of addressing multiple safety and security aspects simultaneously.

To achieve these goals, the proposed system will employ an ESP32 microcontroller as the central processing unit, along with a SIM7600 module for 4G communication and GPS location tracking. Various sensors, including a DHT22 for temperature and humidity, an MQ-5 for gas detection, a PIR sensor for motion detection, and an INA226 power monitor for voltage and current measurement, will be integrated into the system. The entire setup will be powered by a solar-charged battery pack, ensuring sustainability and long-term functionality. In summary, this



General Introduction

boat monitoring system seeks to provide a reliable, energy-efficient, and real-time solution to the challenges faced by boat owners. By leveraging IoT technologies and renewable energy, the system aims to offer an innovative approach to enhancing maritime safety and operational efficiency.

This report mentions different steps to reach the project goals. It is mainly made of three chapters:

The first chapter is **"General Context of The Project"**

The second chapter is **"Hardware And Software Environment"**

The third chapter is **"System Design And Implementation"**

And finally, I am going to conclude all that work and the achieved results with some perspectives that will help us to make this project better and more developed.

Chapter 1

General project context

1.1 Introduction

In this chapter, I will put this project into its general framework and I will start with the general context of it, then I will present the problems and the proposed solutions with a deep description of the project management method, and finally I will conclude this chapter with a preliminary study.

1.2 General Context of the Project

1.2.1 Project scope

In order to deserve a respectful degree of Information and Communication Technologies (ICT) at the Higher Institute of Computer Science and Mathematics in Monastir (ISIMM) in the current academic year of 2024/2025, I did my end-of-studies training at **Hendrik Thureau Enterprises**. The goal of the project is to develop a Boat Monitoring System linked with ‘Boaty Pro’ mobile application. This system aims to provide real-time boat status updates, including location tracking, battery monitoring, environmental conditions, and security alerts. The solution will leverage IoT technologies to enhance maritime safety and operational efficiency for boat owners.

1.2.2 About Hendrick Thureau Enterprises (HTE)

Hendrik Thureau Enterprises (HTE) is a Switzerland-based company specializing in IT and digital business solutions. With business roots dating back to 1988, HTE has built a strong reputation for delivering high-quality digital services, including web, app, and software development, outsourcing, offshoring, digital consulting, and design. The company’s mission is to provide innovative solutions that give clients a competitive edge, enabling their businesses to scale and succeed. Backed by a highly qualified team, HTE is committed to excellence in software development, digital transformation, and strategic business growth.



Figure 1.1: HTE Logo[?]

1.2.3 Problematic

In the maritime industry, boat owners and operators often face challenges in monitoring their vessels remotely, especially when they are docked or left unattended for extended periods. Traditional monitoring methods rely on manual checks or expensive tracking systems, which may not provide real-time insights into a boat's status.

Key issues include :

- **Lack of remote monitoring:** Owners cannot easily check their boat's location, battery health, or onboard conditions without being physically present.
- **Security risks:** Unauthorized access or theft can occur without timely alerts or tracking mechanisms.
- **Environmental hazards:** Sudden changes in temperature, humidity, or gas leaks (such as fuel vapor) can pose safety risks if not detected early.
- **Power management challenges:** Many monitoring systems require constant power sources, making them impractical for small boats with limited energy availability.

To address these challenges, this project aims to develop a Boat Monitoring System integrated with the Boaty Pro mobile application. The system will leverage IoT technologies to provide real-time data on boat location, battery status, environmental conditions, and security alerts, ensuring improved safety, efficiency, and peace of mind for boat owners.

1.2.4 Proposed Solution

To address the challenges identified in the problem statement, this project proposes the development of a Boat Monitoring System integrated with the Boaty Pro mobile application. The system leverages IoT technologies to provide real-time monitoring and data insights, ensuring boat owners can remotely track their vessel's status and receive alerts when necessary.

The solution consists of several key components. It enables remote communication and data transmission over a mobile network, allowing boat owners to access information on demand. The system provides real-time and on-demand location updates to help owners monitor their boat's position. To prevent unexpected power failures, it includes a power monitoring feature that tracks battery voltage, current, and overall consumption. Environmental sensors measure temperature and humidity, ensuring safety against extreme conditions, while gas leak detection helps identify fuel or gas leaks, reducing fire hazards. Security is enhanced with motion detection, which alerts owners to unauthorized access. Additionally, the system is designed for energy efficiency, operating on a solar-powered setup with a rechargeable battery, ensuring continuous functionality even in remote areas.



Chapter 1. General project context

Through Boaty Pro, users can conveniently access real-time and historical data, receive security notifications, and monitor critical boat conditions from anywhere. This cost-effective, energy-efficient, and scalable solution enhances maritime safety, operational efficiency, and user convenience.

1.2.5 Goals and tasks

The content that follows is a list of the major tasks and objectives that the project is meant to achieve:

a. Goals

- Develop a reliable IoT-based monitoring system capable of tracking a boat's location, battery status, and environmental conditions.
- Ensure real-time and on-demand data transmission via a 4G-enabled communication module .
- Improve boat security by integrating motion detection and alert notifications through the mobile app.
- Optimize power consumption using a solar-powered battery system to ensure uninterrupted operation.

b. Key Tasks

- **Installing sensors and communication modules:** Setting up environmental, security, and power monitoring sensors, along with communication modules, to ensure data collection and remote accessibility.
- **Collecting and analyzing data:** Monitoring real-time data from the system, including boat location, battery status, environmental conditions, and security alerts, to provide actionable insights.
- **Developing a remote monitoring system:** Integrating the collected data into the Boaty Pro mobile application, allowing users to view real-time and historical information.

1.3 Existing analysis and Related Technologies

This section explores key technologies relevant to the proposed Boat Monitoring System, highlighting their role in enhancing safety, efficiency, and operational reliability.

1.3.1 Internet of things(IOT)

1.3.1.1 Definition

The Internet of Things (IoT) is a network of physical "objects" or terminals that integrate sensors, software, and other technologies in order to communicate with other terminals and systems on the Internet and exchange data with them.

1.3.1.2 IOT architecture

There are many phases in the IoT architecture and they can be changed depending on the circumstances. When speaking about Iot one can define the following. It contains four phases, which are shown in Figure 1.2.

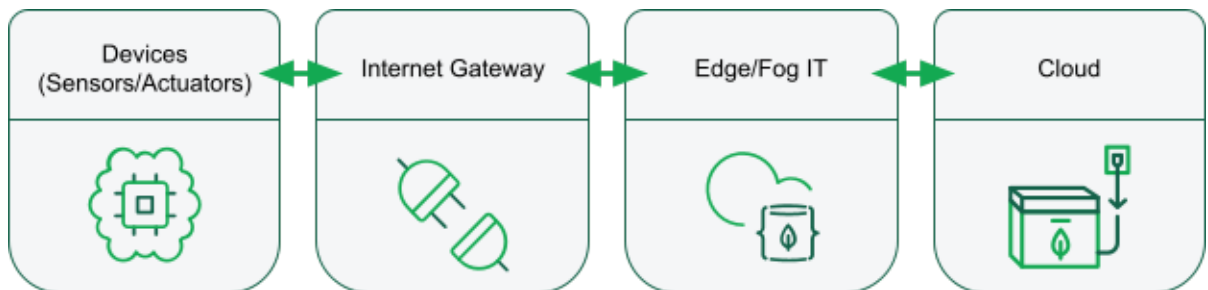


Figure 1.2: Iot architecture[?]

- **Devices:** The Perception layer of IoT focuses on real devices like sensors and actuators that produce data or produce action. The data produced are

Chapter 1. General project context

transformed and sent to the Internet gateway stage. Due to limited resources, data are often transferred in a raw state until a critical decision is made.

- **Internet gateways:** process raw data from devices before sending it to the cloud. It can be a standalone unit or physically linked to the device, connecting to sensors through low-power networks. The gateway can transmit data over the Internet.
- **In edge or fog computing:** data are rapidly processed at the edge of the cloud for quick evaluation and urgent attention. This layer focuses on the latest information required for fast operations, and some pre-processing is carried out to reduce data uploaded to the cloud.
- **Cloud or data center:** In this final stage, the cloud or data center, where data is stored for later processing and deep analysis, including resource-intensive processes such as machine learning training. The saved data can be pushed into dashboards or management software in the application and business layers.[?]

1.3.1.3 IOT application areas

There are several sectors and applications areas where the Internet of Things (IoT) may be used. Among the most popular IoT applications fields are illustrated in the following figure 1.3 :

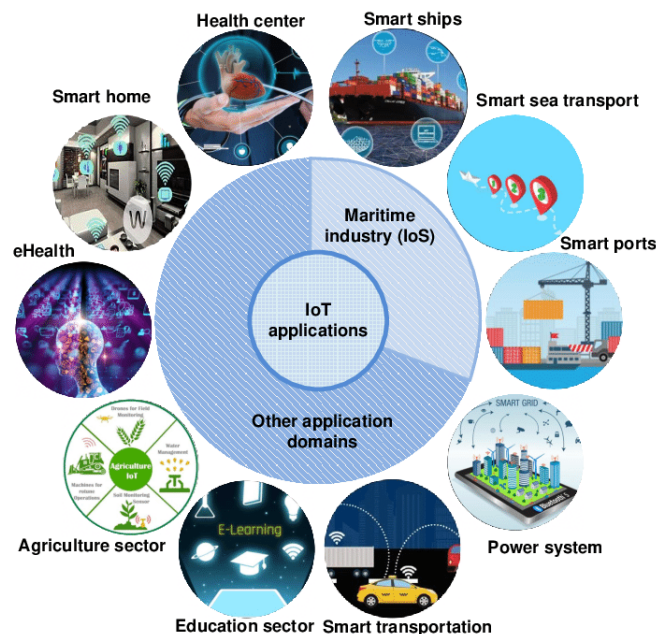


Figure 1.3: Iot application domaines [?]

1.3.2 Embedded systems and IOT

The Internet of Things (IoT) has revolutionized the way we live, work, and interact with each other. IoT refers to a network of interconnected devices that can communicate with each other without human intervention. Embedded systems are at the heart of IoT technology: small and powerful microprocessors embedded in everyday objects such as smartphones, appliances, and vehicles. These systems enable seamless communication between devices, allowing them to collect, process, and share data in real time.

As technologies continue to advance, embedded systems have become more sophisticated and powerful than ever before. They incorporate advanced sensors, wireless capabilities, and machine learning algorithms to deliver highly personalized experiences for users.

By enabling quicker decision-making based on data gathered from various IoT devices, embedded systems are significantly improving our everyday lives in a variety of areas, such as healthcare surveillance, industrial production processes as well as agriculture monitoring.

To sum up, embedded systems are an essential part of the Internet of Things because they supply the required control and intelligence for objects to communicate and share data as seen in figure 1.4 below:

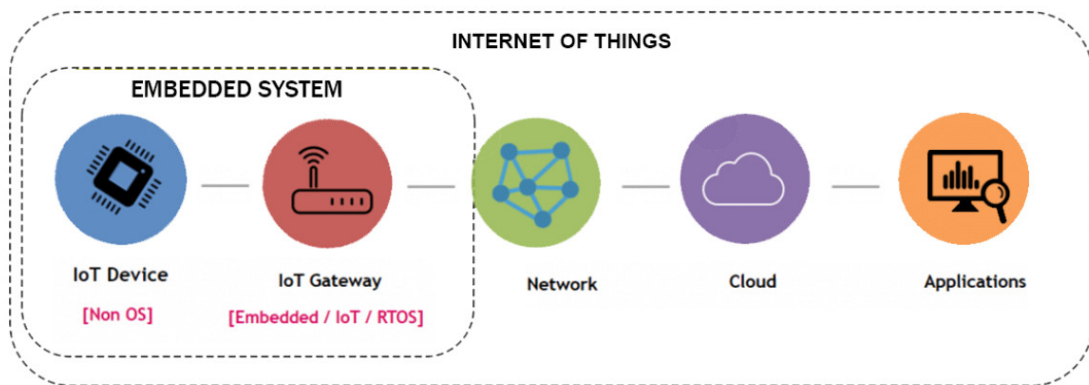


Figure 1.4: An IOT Embedded Systems [?]

1.3.3 Smart Boating

Technology and the IoT impact every aspect of our lives, homes, cars, and lifestyles. So, it is somewhat inevitable that demand for this type of connectivity will eventually reach the boating world too.

Boat owners are beginning to turn to smart technology on their vessels to monitor, track, protect, and enhance their boating experiences. Today's top technologies allow you to connect with and monitor your boat and its systems while you are away. It also allows you to receive up-to-the-minute information while on the water for things such as bilge level alarms or important weather updates.

1.3.3.1 Connected Boating

Connected boating is the overarching term of the use of smart technologies to enhance boat ownership and boating experiences. Just like in other areas of our lives, the boating industry and boat owners are turning to smart technology and connected devices to improve security, monitor systems, and offer real-time updates and alerts.

Specific devices or entire systems can be added to vessels that use sensors to collect data which is then sent to an information hub or cloud, using bluetooth, Wi-Fi or even satellite technology. This information is instantly and continuously analyzed. The information is sent back to the owner via an app subscription, downloaded onto their smartphone or table. In many cases, the owner is able to control and manage devices and switches directly from these apps as well. [<https://waiv.ai>]



Figure 1.5: Connected Boating [?]

1.3.4 The Differences between traditional and Smart Boating

The table 1.1 below presented the main differences between traditional smart (connected) boating:

Chapter 1. General project context

Tableau 1.1: Difference between traditional and smart boating[?]

Traditional	Smart Boating
Manual navigation with maps and compasses.	GPS-based real-time tracking and geofencing.
Requires physical inspection.	Remote monitoring via IoT sensors.
Basic locks and manual surveillance.	Motion sensors, GPS tracking, and alerts.
Manual battery checks and fuel dependency.	Solar-powered system with smart monitoring.

1.3.5 Factors Affecting Boat Performance and Safety

Many factors can significantly impact the performance and safety of a boat. A Smart Boat Monitoring System is designed to monitor and manage these factors in real time, ensuring optimal conditions for safe and efficient operation. Below, we discuss the key factors and their effects:

- **Temperature and Humidity:**

- **Effect:** High temperatures can lead to engine overheating, while high humidity can cause condensation, damaging electrical systems.
- **Monitoring:** Sensors track temperature and humidity levels, generating alerts if thresholds are exceeded.
- **Example:**
 - * Optimal Temperature: 15-30°C
 - * Optimal Humidity: 40-60%
 - * Critical Alert: Temperature > 35°C or Humidity > 80%

- **Power Consumption and Battery Health:**

- **Effect:** Inefficient power management can lead to battery failure, disrupting navigation and communication systems.
- **Monitoring:** Sensors measure voltage, current, and power consumption, providing real-time data and alerts for abnormal conditions.
- **Example:**
 - * Optimal Battery Voltage: 12-14V
 - * Critical Alert: Voltage < 11V or Current > 10A

- **Gas Leaks:**

- **Effect:** Flammable gas leaks pose a serious safety risk, potentially leading to fires or explosions.
- **Monitoring:** Gas sensors detect the presence of flammable gases and trigger alerts if concentrations exceed safe levels.
- **Example:**
 - * Safe Gas Concentration: < 300 ppm
 - * Critical Alert: Gas concentration > 500 ppm
- **Intruder Detection:**
 - **Effect:** Unauthorized access can lead to theft, vandalism, or safety breaches.
 - **Monitoring:** Motion sensors detect movement within the boat's interior and send alerts if motion is detected when the boat is unoccupied.
 - **Example:**
 - * Normal Condition: No motion detected
 - * Intruder Alert: Motion detected when boat is unoccupied
- **Location and Environmental Conditions:**
 - **Effect:** Adverse weather conditions (e.g., high winds, rough seas) and incorrect navigation can compromise safety.
 - **Monitoring:** GPS and environmental sensors provide real-time updates on location, wind speed, wave height, and other conditions.
 - **Example:**
 - * Safe Wind Speed: < 20 knots
 - * High Wave Alert: Wave height > 2 meters
 - * Emergency Alert: Boat drifts outside a predefined safe zone

Figure ?? illustrates how these factors are monitored and managed by the Smart Boat Monitoring System.

1.4 Process for project management

In order to make our project more efficient and successful and to finish it before the deadline, we used methods and tools that were advantageous to us.

1.4.1 Methodologies

The general aim of our project management methodology is to be able to standardize structure and organize work methods.

It helps us reduce risks, avoid duplication of efforts, increase its impact ultimately, resulting in a continuous improvement process. In other words, this methodology is a great tool for generating efficiency.

1.4.1.1 V-model methodology

V- model means Verification and Validation model. Just like the waterfall model, the V-Shaped life cycle is a sequential path of execution of processes. Each phase must be completed before the next phase begins. The following figure 1.7 shows the various phases of a V-Model.[?]

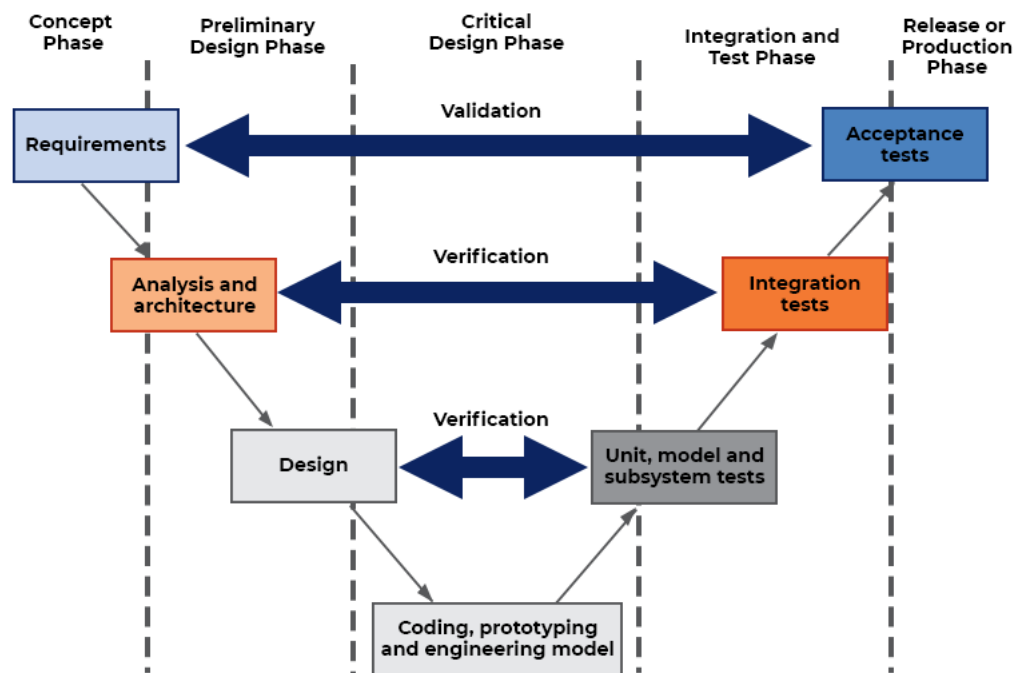


Figure 1.6: V-Model phases[?]

The related testing phase of the development phase is planned in parallel. So, there are Verification phases on one side of the 'V' and Validation phases on the other side. The Coding Phase joins the two sides of the V-Model.

Design Phase

- **Requirements:** This phase contains detailed communication with the customers to understand their requirements and expectations. This stage is known as Requirement Gathering.
- **Analysis and architecture:** Architecture Design: The baseline in selecting the architecture is that it should understand all which typically consists of the list of modules, brief functionality of each module, their interface relationships, dependencies, database tables, architecture diagrams, technology detail, etc.
- **Coding Phase:** After designing, the coding phase started. Depending on the requirements, a suitable programming language is decided. There are some

guidelines and standards for coding, which goes through many code reviews to check the performance.

Testing Phases

- **Unit Testing :** The V-Model uses Unit Test Plans (UTPs) during the module design phase to eliminate errors at the code/unit level. UTPs are executed to verify that the smallest unit (e.g., a program module) can function correctly on its own. Unit testing is essential to ensure the proper functioning of individual program modules.
- **Integration Testing:** Integration Test Plans are developed during the Architectural Design Phase. These tests verify that groups are created and tested independently as well as coexisting and communicating among themselves.
- **Acceptance Testing:** Acceptance testing is done in the user environment to test the software's compatibility and non-functional issues like load and performance defects. It is related to business requirement analysis and reveals any compatibility issues with different systems. This testing is essential to identify any problems in the real user environment.

1.4.2 Project management tools

Organization and scheduling are essential for the success of my project. For managing my work, I used the Bitrix24 platform.

Bitrix24 is a web-based project management tool that helps in job organization and scheduling. It offers various features such as online storage, Scrum, Gantt charts, time management, workload planning, group chat, social networking, knowledge management, video calls, client management, invoicing, document management, file synchronization, HR tools, workflow automation, and customization. It also provides an AI assistant, API integration, and a marketplace for additional tools.

Using Bitrix24 makes it simple to collaborate and communicate by allowing users to manage tasks efficiently, track progress, and streamline workflows. Additionally, its built-in automation and document management features help keep work organized and accessible.

To sum up, Bitrix24 is a powerful and versatile tool that helps individuals and teams stay organized, collaborate effectively, and achieve their goals. The subsequent figures highlight my workspace on this platform.

Chapter 1. General project context

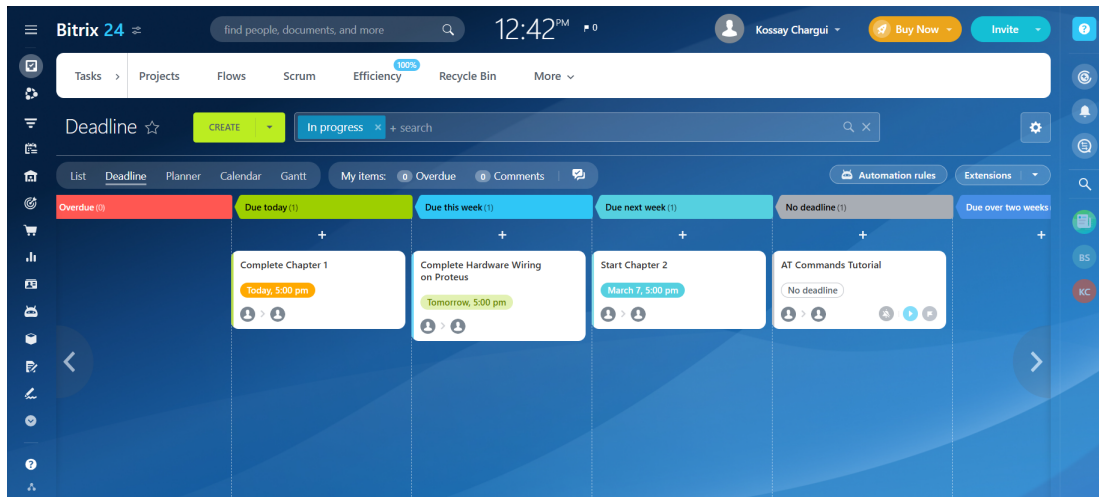
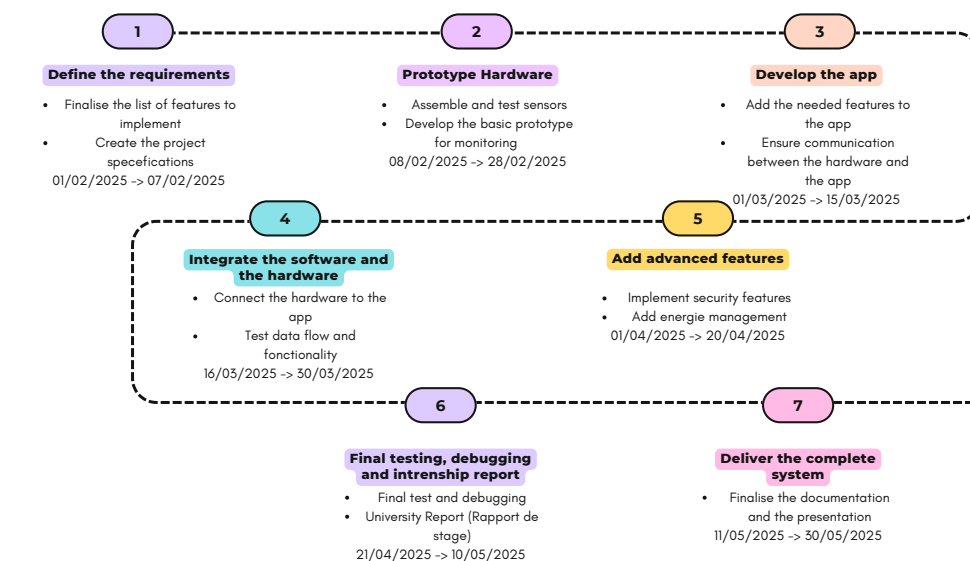


Figure 1.7: Workspace boards

1.4.3 Project planning

In my project, I estimated that it would take approximately four months to complete the work. The timeline provides a detailed overview of the different phases of the project, outlining key milestones and deadlines to ensure a structured and efficient workflow.

Project Timeline



CHARGUI KOSSAY

Figure 1.8: Planning tasks



1.5 Conclusion

In this chapter, I presented the general context of our project, described it, and at the same time we specified the main objectives to be achieved. In the next chapter, a bibliographic analysis of the required materials of the current study will be provided.

Chapter 2

Hardware and software study of the system

2.1 Introduction

In this chapter, I present the hardware and software components chosen for the implementation of the boat monitoring system. Based on the needs identified in the previous chapter, I have selected a set of tools and modules that best match the project's goals in terms of performance, cost, and availability.

The system is built around the LilyGO ESP32 board with an integrated SIM7600 module and GPS for connectivity and location tracking. I also use sensors to detect gas, air quality, motion, temperature, humidity, and monitor battery voltage. This chapter details the rationale behind each component selection and the software tools used for development, forming the technical foundation of the system.

2.2 System solution overview

To address the needs of a reliable and autonomous boat monitoring system, I designed a solution based on a compact and low-power embedded platform capable of collecting, processing, and transmitting sensor data in real time. My goal is to enable remote supervision of various environmental and operational parameters on board, such as gas leaks, air quality, motion detection, temperature, humidity, and battery voltage.

At the heart of the system is the LilyGO TTGO ESP32 development board, which integrates an ESP32 microcontroller, a SIM7600 4G module, and GPS functionality. This all-in-one board significantly reduces complexity, saves space, and simplifies power management. The SIM7600 module allows the system to send data over 4G networks and retrieve GPS location, enabling both monitoring and real-time tracking of the boat.

The sensor suite consists of:

- **DHT22** : for measuring temperature and humidity,
- **MQ-5** : for gas leak detection (e.g., LPG, methane),
- **MQ-135** : for monitoring air quality (e.g., CO, NH),
- **PIR** : sensor for detecting unauthorized movement near or on the boat,
- **INA226** : for voltage and current monitoring of the battery.

All components are connected to the ESP32 via digital or I2C interfaces, depending on the sensor type. The data collected is periodically transmitted to a cloud server or mobile app interface via GSM/4G, allowing me to monitor the system remotely.

Power is supplied by a Li-ion battery in the back of the ESP32, which is charged through a solar panel, ensuring energy autonomy. The system is designed to be energy-efficient, modular, and scalable, with the possibility of integrating additional sensors or actuators in the future.



Chapter 2. Hardware and software study of the system

This solution offers a balance between reliability, cost, and simplicity, making it suitable for real-world marine environments where consistent connectivity and autonomous operation are critical.

2.3 Methodology Approach

2.3.1 Microcontroller Board

The microcontroller board acts as the brain of the system, enabling communication with sensors, modules, and peripherals. In this context, I opted for the LilyGO ESP32 board with a built-in SIM7600 4G module and GPS support. This decision was based on the board's wireless capabilities, low power consumption, and suitability for IoT applications. The table below compares this board with other common microcontroller boards.

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Tableau 2.1: Comparative Study of Microcontroller Boards

Board	Arduino Uno	ESP32 DevKit	LilyGO ESP32 + SIM7600
Dimensions	6.86 cm x 5.34 cm	5.4 cm x 2.7 cm	8.6 cm x 2.7 cm (with SIM7600)
Processor	ATmega328P	Xtensa Dual Core 32-bit LX6	ESP32 + SIM7600 ARM Cortex A7
Memory	32 KB Flash, 2 KB SRAM	4 MB Flash, 520 KB SRAM	4 MB Flash (ESP32), 1 GB RAM (SIM7600)
I/O Pins	14 digital, 6 analog	30 digital, 18 analog	24 GPIO (ESP32), SIM7600 UART interface
Operating Voltage	5V, 3.3V	3.3V	3.3V (ESP32), 3.8-4.2V (SIM7600)
Wireless	None	Wi-Fi, Bluetooth v4.2	Wi-Fi, Bluetooth, 4G LTE, GPS
Clock Speed	16 MHz	80-240 MHz	240 MHz (ESP32) + 1.2 GHz (SIM7600)
Price	41.000 TND	34.800 TND	150.000 TND

2.3.2 Communication protocols

A communication protocol is a set of rules and conventions that govern the format, timing, sequencing, and error control of data exchange between two or more entities. It defines how information is transmitted, received, and interpreted during communication to ensure reliable and consistent data transfer.

In my project, communication protocols play a crucial role in enabling seamless interaction between onboard sensors, control units, and the cloud-based monitoring platform. The main objective is to ensure real-time or periodic transmission of location, environmental, and system status data from the boat to the end user, even when the boat is in remote areas with limited or no terrestrial connectivity.

As can be seen in figure ??, there are different types of wireless networks according to their range and bandwidth.

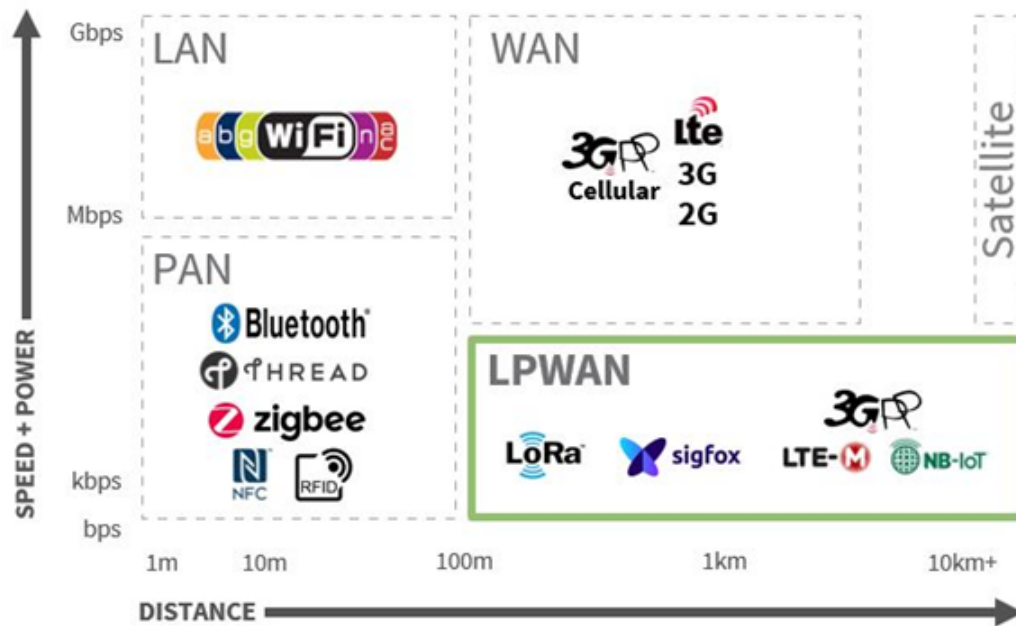


Figure 2.1: Range and bandwidth for different types of wireless networks [?]

There are two primary categories of networks available in the market: long-range networks and short-range networks.

- **Long-range networks:** including LPWAN technologies like Sigfox, LoRa, and cellular networks (GSM, 2G, 3G, 4G, 5G), facilitate data transfer over extensive distances. They are widely employed to connect infrastructure spread over kilometers, offering reliable long-range communication capabilities.
- **Short-range networks:** like WiFi, Z-Wave, ZigBee, and low-energy Bluetooth, facilitate data transfer over shorter distances. They are widely used in home automation and consumer wearables for reliable local communication.

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Attribute	Bluetooth® Low Energy Technology	Wi-Fi	Z-Wave	IEEE 802.15.4 (Zigbee, Thread)	LTE-M	NB-IoT	Sigfox	LoRaWAN
Range	10 m – 1.5 km	15 m – 100 m	30 m – 50 m	30 m – 100 m	1 km – 10 km	1 km – 10 km	3 km – 50 km	2 km – 20 km
Throughput	125 kbps – 2 Mbps	54 Mbps – 1.3 Gbps	10 kbps – 100 kbps	20 kbps – 250 kbps	Up to 1 Mbps	Up to 200 kbps	Up to 100 bps	10 kbps – 50 kbps
Power Consumption	Low	Medium	Low	Low	Medium	Low	Low	Low
Ongoing Cost	One-time	One-time	One-time	One-time	Recurring	Recurring	Recurring	One-time
Module Cost	Under \$5	Under \$10	Under \$10	\$8-\$15	\$8-\$20	\$8-\$20	Under \$5	\$8-\$15
Topology	P2P, Star, Mesh, Broadcast	Star, Mesh	Mesh	Mesh	Star	Star	Star	Star
Shipments in 2019 (millions)	~3,500	~3,200	~120	~420	~7	~16	~10	~45

Figure 2.2: Comparison of different wireless communication protocols

While short-range technologies like Bluetooth and WiFi may perform well in indoor or terrestrial environments, they become unreliable or unusable in maritime environments. This is due to their limited range and dependence on local infrastructure, such as access points or routers—which are simply unavailable in open sea conditions. Similarly, LoRaWAN requires the deployment of a gateway to collect and transmit data from the nodes, meaning two LoRa modules are needed: one in the remote system and one acting as the base station. This infrastructure dependency makes LoRaWAN infeasible for mobile and isolated use cases like our boat monitoring system.

Moreover, traditional satellite communication systems, such as those offered by Iridium or Inmarsat, are already used in maritime applications for emergency signaling and remote data exchange. However, these systems are notoriously expensive, both in terms of hardware and operational costs. They typically require specialized, high-gain antennas and bulky modems, which are not suitable for compact, low-power IoT systems like the one proposed in our project. This makes them inaccessible for many small-scale applications or budget-constrained deployments.

By contrast, the emerging Direct to Cell technology from Starlink provides a cost-effective and scalable alternative. It leverages existing LTE protocols and is designed to work with unmodified mobile devices and modules, eliminating the need for expensive satellite-specific hardware. This approach dramatically lowers the entry barrier for global connectivity, especially in hard-to-reach environments like the sea.

After comparing the different types of networks, we finally decided to use a cellular network-based solution. Our system takes inspiration from the Starlink “Direct to Cell” innovation, which enables communication between satellites and standard LTE devices. This solution offers a robust, future-proof infrastructure that is expected to support not only messaging but also voice, data, and IoT services

by 2025. As documented by Starlink in their official update [?], this technology overcomes the significant technical challenges of satellite-to-phone communication by incorporating phased-array antennas, custom silicon, and advanced signal processing onboard the satellites.

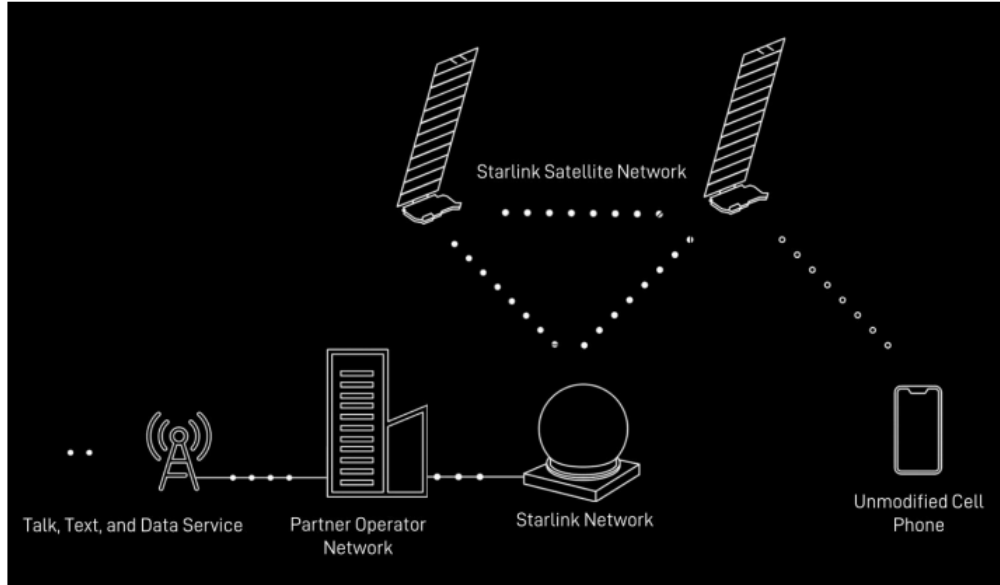


Figure 2.3: Starlink Direct to Cell concept: satellites act as LTE towers in space [?]

2.4 Hardware Section

This section provides an in-depth analysis and technical study of the hardware components chosen for the boat monitoring system. The selection of each component was based on its suitability, performance, energy efficiency, and cost-effectiveness to meet the system requirements for remote monitoring, GPS tracking, and environmental sensing in marine environments.

The proposed hardware system is composed of the following main components:

- ESP32 development board with SIM7600 4G/GPS module
- Sensors (PIR motion, MQ-5 gas, MQ-135 air quality, DHT22 temperature/humidity, INA226 voltage/current)
- Power supply system (single 18650 Li-ion battery, solar panel input, onboard charger)
- Supporting modules (voltage regulators, external antennas, connectors)

2.4.1 ESP32 with SIM7600 Module

The core of the system is the LilyGO TTGO ESP32 board equipped with an ESP-WROVER-E microcontroller and integrated SIM7600 4G LTE/GNSS module. Key specifications from the ESP-WROVER-E datasheet include:

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- **Processor:** Xtensa dual-core 32-bit LX6 microprocessor (up to 240 MHz)
- **Memory:**
 - 448 KB ROM
 - 520 KB SRAM + 16 KB RTC SRAM
 - 4/8/16 MB SPI flash (configurable)
 - 8 MB PSRAM (ESP32-D0WDR2-V3 variant)
- **Wireless:**
 - Wi-Fi 802.11b/g/n (150 Mbps max)
 - Bluetooth 4.2 BR/EDR + BLE
- **Peripherals:** 24 GPIO pins supporting UART, SPI, I2C, PWM, ADC, and CAN bus
- **Power:** 3.0–3.6 V operating range, optimized for low-power sleep modes

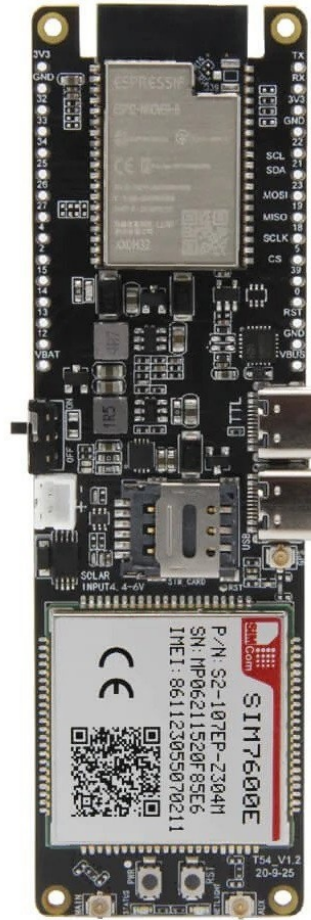


Figure 2.4: LilyGO TTGO ESP32 (ESP-WROVER-E variant) with integrated SIM7600 module. Source: LilyGO GitHub repository [?].

2.4.1.1 Key Advantages for Marine Applications

- **Enhanced Memory:** 8MB PSRAM enables robust buffering of GPS/Cellular data during intermittent connectivity.
- **Dual-Core Efficiency:** Dedicate one core to sensor polling (DHT22, INA226) while the other manages cellular/GNSS communications.
- **Industrial-Grade Reliability:** Operates at -40°C to 85°C , critical for harsh marine environments.
- **Integrated Antenna Options:** On-board PCB antenna (WROVER-E) or external connector (WROVER-IE) for flexible deployment.

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Tableau 2.2: ESP-WROVER-E Key Specifications (Source: Espressif Datasheet [?])

Parameter	Value
Cellular Modem	SIM7600E-H (LTE Cat-1, up to 150 Mbps DL)
GNSS Support	GPS, GLONASS, Galileo, BeiDou
PSRAM	8 MB (ESP32-D0WD-R2-V3)
Deep Sleep Current	1.2 mA
Certifications	Bluetooth BQB, RoHS, REACH

a. LilyGO TTGO ESP32 Pinout

The LilyGO TTGO ESP32 board integrates the ESP32 microcontroller and SIM7600 module on a compact development board. All GPIO, power, and communication lines are exposed via pin headers along both edges, facilitating straightforward connection of sensors, antennas, and power circuits with jumper wires or header cables. This layout allows secure, reliable interfacing even in a marine enclosure.

Figure 2.5 illustrates the pin layout of the TTGO ESP32 + SIM7600 board, highlighting the 3.3 V, GND, I2C (SDA/SCL), UART (TX/RX), and analog input pins used for sensor integration.

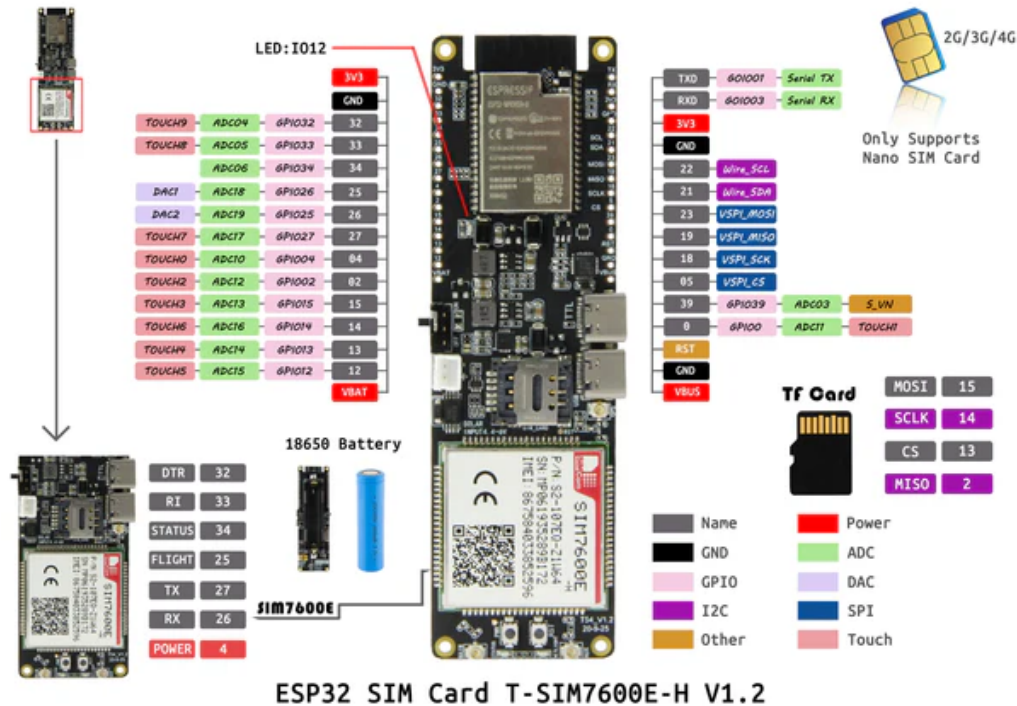


Figure 2.5: LilyGO TTGO ESP32 + SIM7600 Pinout

b. ESP32 Programming Guide

The ESP32 on the LilyGO board can be programmed using several frameworks and languages, offering flexibility depending on project requirements:

- Arduino C/C++ (via the ESP32 Arduino core)
- Espressif IoT Development Framework (ESP-IDF)
- MicroPython
- JavaScript (using Espruino or Moddable SDK)

For rapid prototyping and access to a rich ecosystem of libraries (SIM7600 AT command wrappers, TinyGPS++ for GNSS parsing, and sensor drivers), I use the Arduino C/C++ environment. This approach simplifies serial debugging, deep-sleep configuration, and OTA updates when connected via USB or 4G.

2.4.2 SIM7600 4G LTE & GNSS Module

The SIM7600E-H is a multi-band LTE Cat-1 cellular modem with integrated GNSS support, chosen for its combination of high-speed data, global satellite navigation, and ease of integration with the ESP32. It provides reliable 4G connectivity for real-time data transmission and multi-constellation positioning (GPS, BeiDou, GLONASS, Galileo) without the need for separate modules.



Figure 2.6: SIM7600E-H LTE GNSS Module (Source: Xinyuan-LilyGO)

From a technical standpoint, the SIM7600E-H module features:

- **LTE Cat-1 Data:** Up to 10 Mbps uplink, 50 Mbps downlink
- **GNSS:** Concurrent GPS, BeiDou, GLONASS, Galileo
- **Frequency Bands:** LTE: B1/B2/B3/B4/B5/B8/B12/B13/B18/B19/B20/B25/B26/B27/B28
UMTS: B1/B2/B5/B8; GSM: 850/900/1800/1900 MHz
- **Interfaces:** UART (for AT commands), USB 2.0, I²C, PCM
- **Power Supply:** 3.4 – 4.2 V (Li-ion); up to 2 A peak during transmission
- **SIM Card:** Standard SIM socket, supports 1.8 V/3 V cards
- **Antenna Connectors:** SMA for LTE and GNSS antennas

a. Technical Specifications

Table 2.3 lists the key specifications of the SIM7600E-H module as used on the LilyGO TTGO board.

Tableau 2.3: SIM7600E-H Technical Specifications

Parameter	Value
Supply Voltage	3.4 – 4.2 V
Average Current (idle)	< 25 mA
Peak Current (LTE transmit)	Up to 2 A
Data Rates	UL: up to 10 Mbps; DL: up to 50 Mbps
Module Dimensions	42 mm × 36 mm × 4.2 mm
Cellular Bands	LTE, UMTS, GSM (see bullet list above)
GNSS Channels	32 parallel satellite channels
Operating Temperature	−40 °C to +85 °C
Interfaces	UART, USB 2.0, I ² C, PCM
Certifications	CE, FCC, PTCRB (module)

b. Pinout Description

Figure 2.7 shows the pin assignments for the SIM7600E-H module as mounted on the LilyGO board. Table 2.4 describes each pin's function.

Chapter 2. Hardware and software study of the system

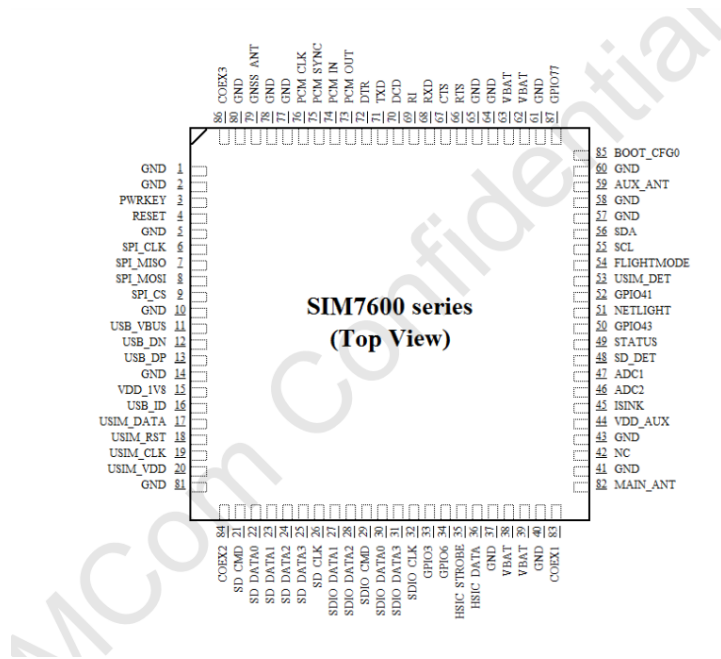


Figure 2.7: SIM7600E-H Pinout Diagram (annotated)

Tableau 2.4: SIM7600E-H Pinout Description

Pin	Function
VCC	Power input (3.4 – 4.2 V Li-ion battery)
GND	Ground reference
TXD	UART transmit (module → MCU)
RXD	UART receive (MCU → module)
USB_D+ / USB_D	USB data lines (for flashing/debug)
I2C_SDA / SCL	I ² C data and clock (optional debugging)
PCM_CLK / DAT	Digital audio interface (PCM)
NET_LED	Network status indicator LED
RESET	Active-low reset input
DTR	Sleep control (HIGH: sleep; LOW for > 50 ms: wake)
RTC_RST	Wake-up from deep sleep
SIM_DET	SIM card detection (MICRO SIM socket present)

2.4.3 Sensors

Sensors in the boat monitoring system measure environmental and operational parameters to ensure safety and performance at sea. They collect data on temperature, humidity, air quality, gas presence, motion, and battery status, which is then transmitted via the ESP32+SIM7600 module to the remote monitoring platform.

2.4.3.1 DHT22 Sensor

The DHT22 sensor (also known as AM2302) uses a capacitive humidity sensing element and a thermistor to measure humidity and temperature, respectively. It employs a single-wire digital communication protocol: after a start signal from the ESP32, the DHT22 sends a 40-bit data packet (16 bits humidity, 16 bits temperature, 8 bits checksum) over one data line. A full data exchange takes approximately 5 ms.

Communication and synchronization between the ESP32 and the DHT22 occur on a single GPIO pin. The microcontroller pulls the data line low for at least 1 ms to initiate a read, then listens for the sensor's 40 bit response.

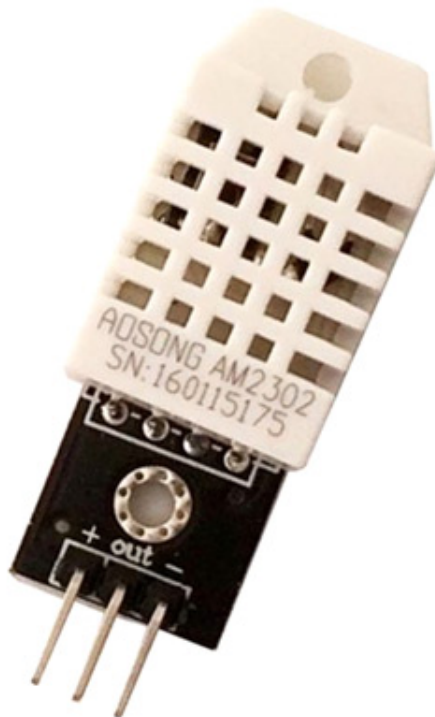


Figure 2.8: DHT22 Temperature and Humidity Sensor Module[1]

The DHT22 sensor offers the following features:

- Temperature range: $-40\text{ }^{\circ}\text{C}$ to $+80\text{ }^{\circ}\text{C}$ with $\pm 0.5\text{ }^{\circ}\text{C}$ accuracy

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- Humidity range: 0 to 100% RH with $\pm 2\%$ accuracy
- Sampling rate: 0.5 Hz (one reading every 2 seconds)
- Operating voltage: 3.3 – 6 V
- Low power consumption: typically 1.5 mA during measurement

a. Technical Specifications

Tableau 2.5: DHT22 Technical Specifications[16]

Parameter	Value
Dimensions	15.1 mm $\times$ 25 mm $\times$ 7 mm
Power Supply	3.3 – 6 V
Average Current	1.5 mA (during measurement)
Temperature Measurement	-40 °C to +80 °C, 0.1 °C resolution, ± 0.5 °C accuracy
Humidity Measurement	0 % to 100 % RH, 0.1% resolution, ± 2 % accuracy
Sampling Rate	0.5 Hz (one reading every 2 s)
Response Time	< 2 s
Price	14.500TND

b. Pinout Description

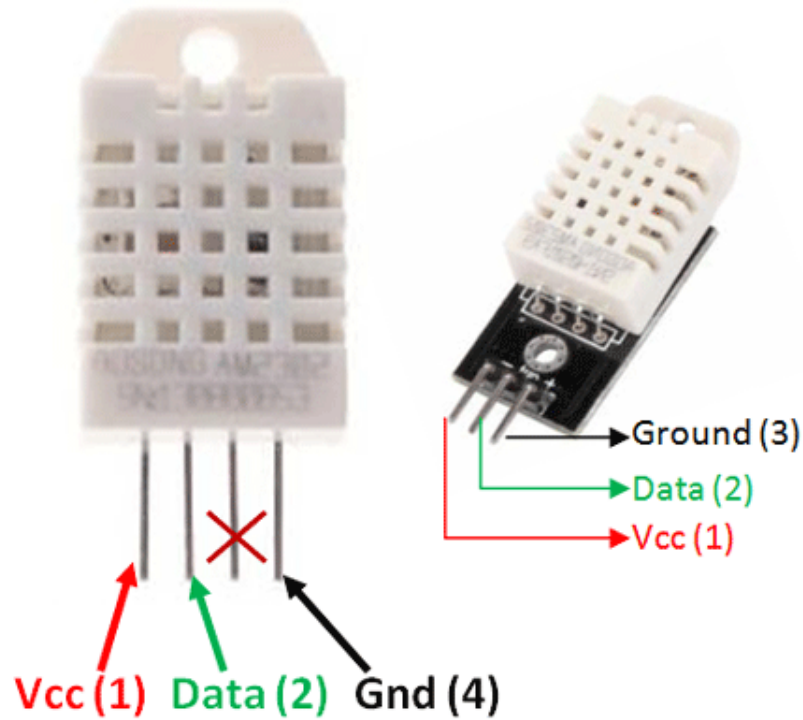


Figure 2.9: DHT22 Module Pinout[2]

Tableau 2.6: DHT22 Pinout Description[2]

Pin	Function
VCC	Power supply (3.3 – 6 V)
DATA	Single-wire digital data I/O
NC	Not connected
GND	Ground

2.4.3.2 PIR Motion Sensor

The PIR (Passive Infrared) motion sensor (HC-SR501) detects changes in infrared radiation within its field of view, allowing it to sense the presence or movement of humans or animals near the boat. Internally, it uses a pyro-electric sensor to generate a small voltage when it detects infrared heat variations. The sensor board includes a Fresnel lens, adjustable sensitivity and timing potentiometers, and outputs a digital HIGH signal when motion is detected.

Upon power-up, the PIR sensor enters a warm-up period (typically 30–60 s). After stabilization, the sensor continuously monitors infrared levels. When motion is detected, the digital output pin goes HIGH for a duration set by the

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onboard “time” potentiometer (0.3–18 s). Sensitivity can be adjusted via the “sensitivity” potentiometer or by selecting one of two detection ranges via the onboard jumper (3 m or 7 m maximum).



Figure 2.10: HC-SR501 PIR Motion Sensor Module[3]

The PIR sensor offers the following features:

- Detection range: adjustable, typically 3–7 m
- Field of view: 120°
- Operating voltage: 3.3–5 V
- Low standby current: $<50 \mu\text{A}$
- Onboard potentiometers for sensitivity and timing adjustment

a. Technical Specifications

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Tableau 2.7: HC-SR501 PIR Sensor Technical Specifications[17]

Parameter	Value
Model	HC-SR501
Detection Distance	3–7 m (adjustable)
Detection Angle	120°
Trigger Mode	Repeatable (RET), Non-repeatable (L)
High Level Output	3.3–5 V
Quiescent Current	<50 μ A
Operating Voltage	3.3–5 V
Warm-up Time	30–60 s
Time Delay	0.3–18 s (adjustable)
Operating Temperature	–15 °C to +70 °C
Module Dimensions	32 mm $\times$ 24 mm

b. Pinout Description

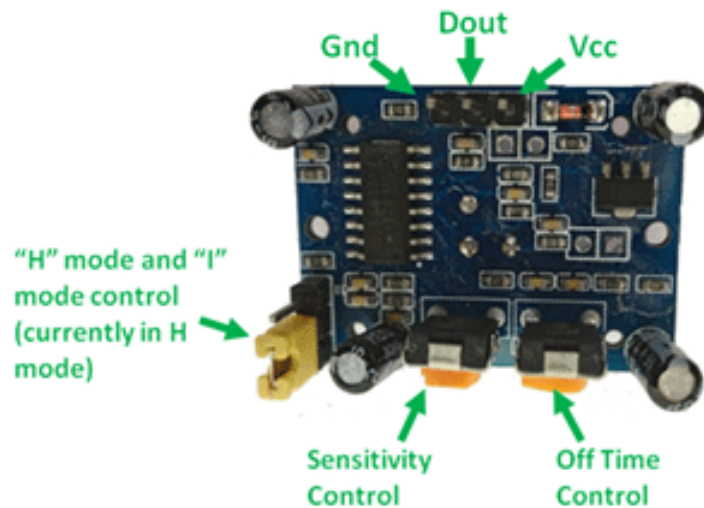


Figure 2.11: HC-SR501 PIR Sensor Pinout[4]

Tableau 2.8: HC-SR501 PIR Sensor Pinout Description[4]

Pin	Function
VCC	Power supply (3.3–5 V)
OUT	Digital output: HIGH when motion is detected
GND	Ground

2.4.3.3 INA226 Power Monitoring Sensor

The INA226 is a high-side or low-side I2C-based current and power monitoring sensor, used in our system to measure the battery's voltage, current, and power consumption accurately. It integrates a 16-bit analog-to-digital converter (ADC) and performs computations internally, outputting digital values over the I2C bus.

The sensor requires a shunt resistor placed in series with the load. The INA226 measures the voltage drop across this resistor to calculate the current, while also monitoring the bus voltage to determine power consumption. Its ability to monitor both voltage and current makes it suitable for low-power embedded systems and energy monitoring projects.

The figure 2.12 presents the INA226 module utilized in our project.

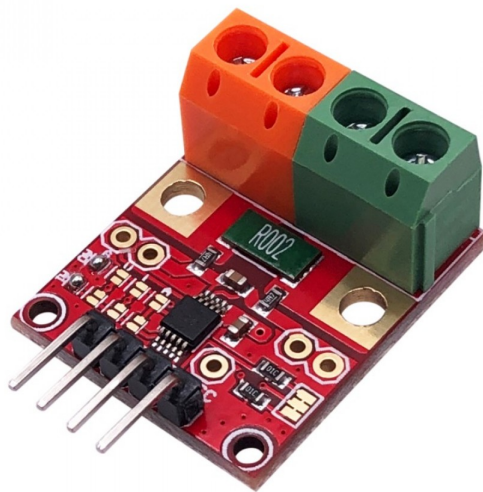


Figure 2.12: INA226 Power Monitoring Module[5]

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a. Technical Specifications

The table below summarizes the key specifications of the INA226 sensor:

Tableau 2.9: INA226 Technical Specifications[18]

Operating Voltage	2.7V – 5.5V
Bus Voltage Range	0V – 36V
Current Measurement Range	$\pm 15\text{A}$ (depends on shunt resistor)
Communication Interface	I ² C (up to 400kHz)
ADC Resolution	16-bit
Power Consumption	Low (typically <1mA)
Dimensions	25mm × 21mm
Price	15.000 TND

b. Pinout Description

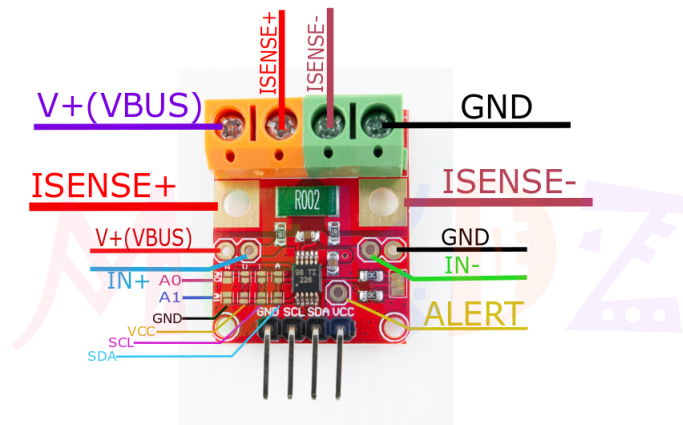


Figure 2.13: INA226 Pinout Diagram[6]

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Tableau 2.10: INA226 Pinout Description[6]

Pin	Description
VCC	Power supply input (2.7V–5.5V).
GND	Ground connection.
SDA	I ² C data line for communication with the microcontroller.
SCL	I ² C clock line for communication synchronization.
VIN+	Positive input for current sensing (connect to power source side of shunt resistor).
VIN–	Negative input for current sensing (connect to load side of shunt resistor).

2.4.3.4 Gas and Air Quality Sensors

In the context of our boat monitoring system, gas detection and air quality assessment are crucial for safety and environmental awareness. We employed two gas sensors:

- The **MQ-5** sensor to detect combustible gases such as LPG, methane, hydrogen, and natural gas, providing real-time gas leakage alerts.
- The **MQ-135** sensor to monitor air quality by measuring concentrations of harmful gases like ammonia, nitrogen oxides, benzene, and smoke, thus indicating the overall air pollution level in the boat environment.

a. MQ-5 Gas Sensor (Combustible Gas Detection) The MQ-5 sensor is a sensitive semiconductor-based gas sensor designed to detect combustible gases. It consists of a gas-sensitive layer formed on a ceramic substrate. When exposed to gas, the sensor's conductivity changes, allowing it to detect gas concentrations based on voltage variations across its output.

The sensor outputs an analog signal and can also be used with a digital threshold-based output through a comparator. Figure 2.14 shows the MQ-5 module.



Figure 2.14: MQ-5 Gas Sensor Module[7]

Technical Specifications

Tableau 2.11: MQ-5 Technical Specifications[19]

Operating Voltage	5V DC
Detection Range	300 – 10000 ppm (LPG, methane, hydrogen)
Preheat Time	Not less than 48 hours
Analog/Digital Output	AO (analog), DO (digital via comparator)
Dimensions	32mm × 20mm × 22mm
Price	10.000 TND

Pinout Description

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Tableau 2.12: MQ-5 Pinout Description[20]

Pin	Description
VCC	Power supply (5V)
GND	Ground
AO	Analog output signal proportional to gas concentration
DO	Digital output signal based on comparator threshold

b. MQ-135 Air Quality Sensor The MQ-135 sensor is suitable for detecting a wide range of gases associated with poor air quality. It is widely used for monitoring indoor air pollution levels. This sensor is sensitive to gases such as ammonia (NH_3), nitrogen oxides (NO_x), alcohol, benzene, smoke, and carbon dioxide (CO_2).

The MQ-135 works similarly to the MQ-5: it has a sensitive layer whose resistance varies with gas concentration. Figure 2.15 shows the MQ-135 module.



Figure 2.15: MQ-135 Air Quality Sensor Module[8]

Technical Specifications

Tableau 2.13: MQ-135 Technical Specifications[21]

Operating Voltage	5V DC
Detection Range	10 – 1000 ppm (various gases : NH ₃ ,NO _x , alcohol, Benzene, smoke,CO ₂ ,etc.)
Preheat Time	Over 24 hour
Analog/Digital Output	AO (analog), DO (digital via comparator)
Dimensions	32mm × 20mm × 22mm
Price	11.000 TND

Pinout Description

Tableau 2.14: MQ-135 Pinout Description[22]

Pin	Description
VCC	Power supply (5V)
GND	Ground
AO	Analog output signal proportional to pollution level
DO	Digital output signal based on threshold setting

2.4.4 Power Sources

In this section, I provide a detailed description of the power sources utilized in the boat monitoring system.

2.4.4.1 18650 Li-ion Cell

The primary power source is a single 18650 lithium-ion cell, which operates at a nominal voltage of 3.7V and is housed in the onboard battery holder of the LilyGO TTGO ESP32 SIM7600 board. This cell offers high energy density, rechargeability, and built-in protection against over-charge and over-discharge via the board's CN3165 charging IC.



Figure 2.16: 18650 Li-ion Cell in Battery Holder[9]

Description

- Nominal Capacity: 2600mAh
- Usable Capacity: 2200mAh (85
- Nominal Voltage: 3.7V (operating range 3.4–4.2V)
- Maximum Charge Voltage: 4.2V
- Operating Temperature: Charge 0°C–45°C, Discharge –20°C–60°C
- Estimated Runtime: 11h at 200mA average draw
- Price: 18.000TND

2.4.4.2 Solar Panel

To achieve energy autonomy, I plan to connect a 6V/10W solar panel (340×220mm) directly to the board's solar input. The CN3165 charge controller accepts up to 6V, regulating current (up to 500mA) to recharge the 18650 cell during daylight hours.



Figure 2.17: 6V/10W Solar Panel (340×220mm)[10]

Description

- Rated Voltage: 6V (open-circuit)
- Rated Power: 10W
- Rated Current: 1.67A (at peak sun)
- Dimensions: 340×220mm
- Features: UV-resistant, corrosion-proof backing, tempered glass
- Expected Daily Yield: 18Wh (5h full sun)
- Price: 25.000TND

2.4.5 Reliability in Maritime Environments

Operating in marine conditions exposes the monitoring system to salt spray, high humidity, wide temperature fluctuations, UV radiation, vibration, and biofouling. To ensure long-term reliability, I have evaluated and mitigated environmental risks for each component: the ESP32+SIM7600 board and INA226 sensor are conformally coated and fully enclosed; the DHT22 sits under a ventilated, UV-stable radiation shield; MQ-5 and MQ-135 gas sensors use replaceable cartridges behind hydrophobic membranes for corrosion protection and in-field calibration; the PIR module is fitted with a hydrophobic lens cover and tuned for reduced sensitivity; and cellular/GNSS uses IP-rated external antennas with strain relief. Combined with local data buffering and periodic calibration routines, these measures safeguard uptime and data integrity in harsh offshore deployments.

2.5 Software Section

In this part, I describe the software tools used during development of the boat monitoring system.

2.5.1 Simulation Software

Fritzing is an open-source tool for designing and documenting electronic circuits. It provides schematic, breadboard, and PCB views, allowing drag-and-drop placement of components and virtual wiring to prototype circuit designs before hardware assembly.



Figure 2.18: Fritzing Logo[11]

2.5.2 Computer-Aided Design Software (CAD)

To design the enclosure and PCB layout in 3D, I use **SolidWorks**. It enables precise 3D modelling of mechanical parts and enclosures, ensuring that the electronics fit securely and withstand marine conditions. If requirements change, I can switch to another CAD package such as Autodesk Inventor without altering the workflow.



Figure 2.19: SolidWorks Logo[12]

2.5.3 Integrated Development Environment (IDE)

Arduino IDE is the main development environment used to program the ESP32 board. It provides a simple and beginner-friendly interface for writing, compiling, and uploading embedded code.

Key features include:

- Support for a wide range of microcontrollers including ESP32
- Integrated serial monitor for debugging
- Rich library ecosystem
- Lightweight and easy to use

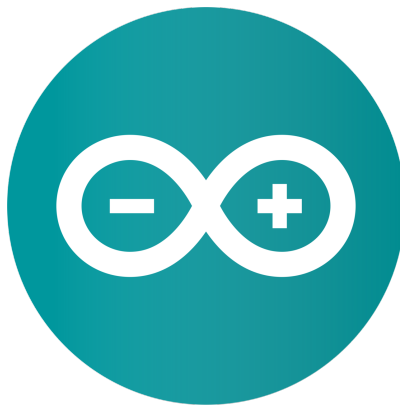


Figure 2.20: Arduino IDE Logo[13]

In addition, I also use **Visual Studio Code** for mobile and web development tasks.

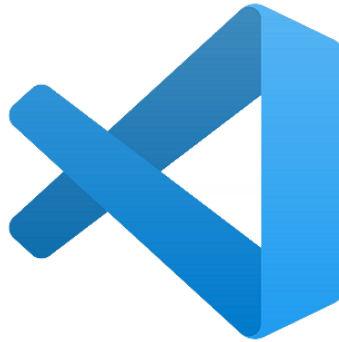


Figure 2.21: Visual Studio Code Logo[14]

2.5.4 Mobile Application Integration

Instead of using a generic IoT platform such as Blynk, I have integrated sensor data and control logic directly into the **Boat Pro** mobile application, developed with **Angular** and **Ionic**. This allows for a customized and seamless user experience.



Figure 2.22: Ionic Framework Logo[15]

Key benefits of this approach include:

- A fully customized interface for boat monitoring
- Integration with maps and charts for location and sensor data
- Support for push notifications and offline access



2.6 Conclusion

In this chapter, I have examined both the hardware and software elements of the boat monitoring system in detail. I described the system architecture, justifying the selection of each hardware component—including the ESP32+SIM7600 board, sensors, and power sources—and outlined the communication protocols suited to maritime conditions. I also reviewed the software environment, from circuit simulation in Fritzing to the 3D CAD tools for enclosure design, as well as the IDEs and frameworks used for firmware and mobile-app integration. These choices form a solid technical foundation for the project’s implementation.

In the next chapter, I will present the step-by-step realization of the prototype, including wiring, PCB design, firmware deployment, and initial testing results, to demonstrate the system’s functionality in real-world conditions.

Chapter 3

Prototype System Implementation

3.1 Introduction

Through the preceding chapters, I have defined the objectives of the boat monitoring system, selected and justified the hardware components, and established the software environment. In this chapter, I move to the implementation phase. I will assemble the embedded hardware, integrate all sensors and communication modules, and deploy the firmware. Then I will configure and test the Boat Pro mobile application to receive and display real-time data from the vessel. This gradual, step-by-step approach—accompanied by screenshots and test cases—demonstrates the successful realization and validation of the complete system.

3.2 Realization and Validation

In this section, I present the results of the system’s realization and validation. First, I detail the wiring and assembly of the ESP32+SIM7600 board, sensors (DHT22, MQ-5, MQ-135, PIR, INA226), and power supply within the marine enclosure. Next, I describe the tests performed to verify GPS/GSM connectivity, sensor accuracy, and power autonomy. Finally, I illustrate the configuration of the Boat Pro Ionic/Angular application and show the live data dashboards and alert mechanisms.

3.2.1 Boat Monitoring System

3.2.1.1 Electronic Schematic

To provide a comprehensive overview, Figure 3.1 shows the global wiring of the boat monitoring system, including power, I²C bus, UART lines to the SIM7600, and sensor connections. Figure 3.2 presents the detailed Fritzing schematic used to verify pin assignments and signal routing.

Chapter 3. Prototype System Implementation

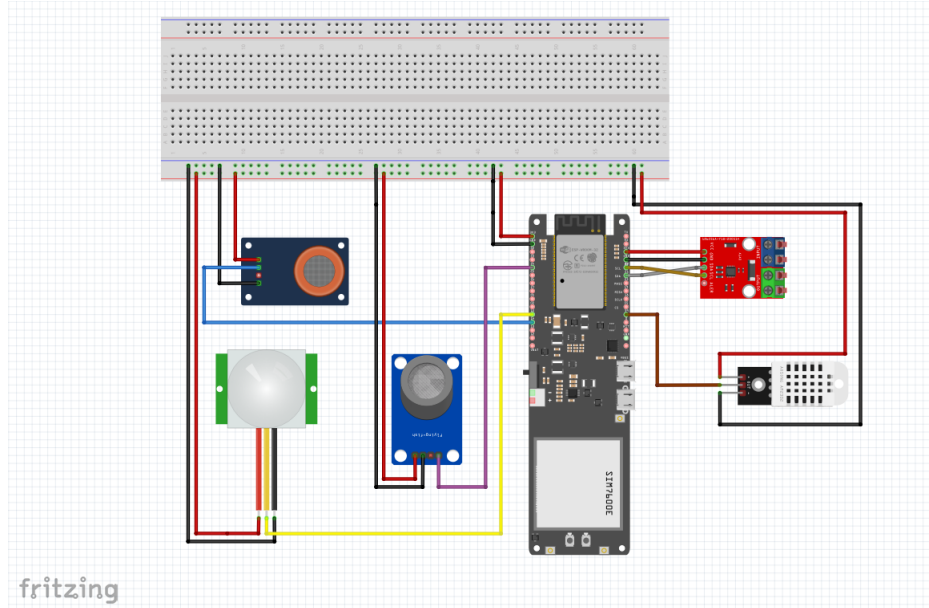


Figure 3.1: Global Wiring Diagram of Boat Monitoring System

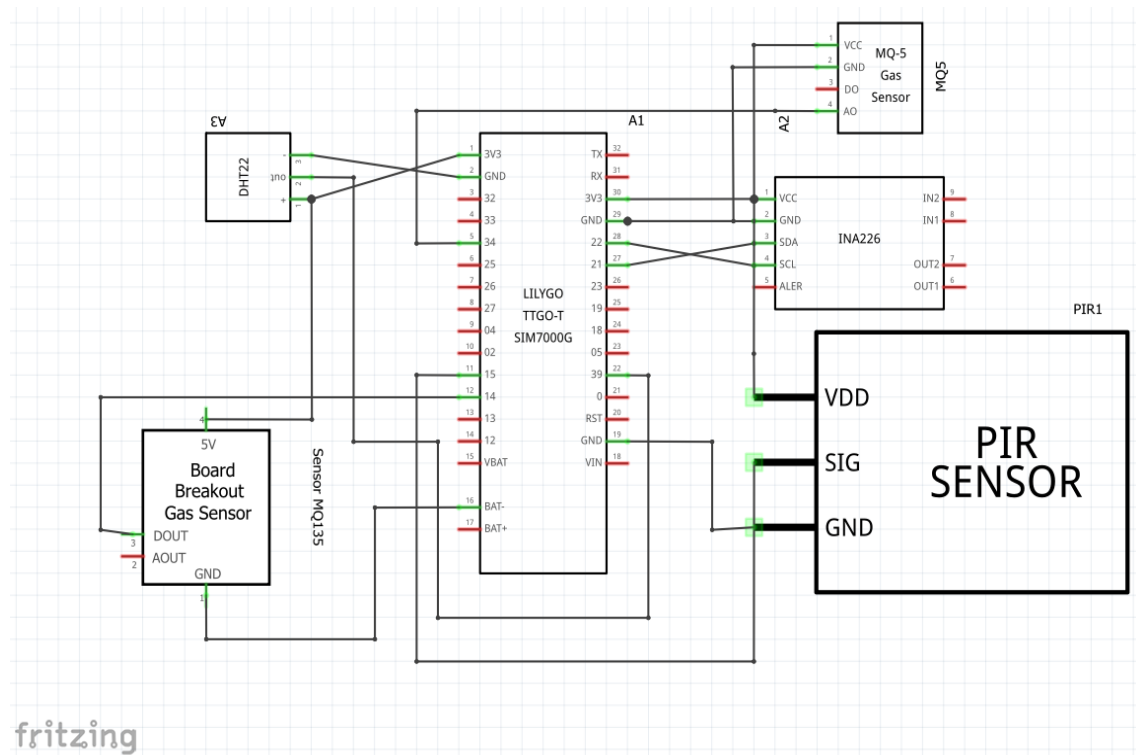


Figure 3.2: Fritzing Schematic of Boat Monitoring System

3.2.2 SIM7600 Initialization & Testing

The built-in SIM7600 modem on the LilyGO TTGO ESP32 board is initialized and configured entirely via AT commands over its UART interface. The sequence below mirrors the commands issued in firmware:



Chapter 3. Prototype System Implementation

1. `AT+CLTS=1` — Enable network time sync (sent via `SerialAT`)
2. `AT+CTZU=1` — Enable automatic time zone update (sent via `modem.sendAT()`)
3. `AT+CGNSSMODE=1,1` — Set GPS-only mode (sent via `modem.sendAT()`)
4. `AT+CGNSS=1` — Enable GPS (via `modem.enableGPS()`)
5. `AT+HTTPSSL=0` — Disable HTTPS (missing in original list, used in `sendHTTPPost()`)
6. `AT+CMGF=1` — SMS text mode (for alerts)
7. `AT+CMGS` — Send SMS alert (with phone number)

These commands verify module health, enable GPS, establish a GPRS context, and transmit sensor readings via HTTP—all without additional external wiring.

3.2.3 Boat Monitoring Prototype

3.2.3.1 Hardware Integration

I mounted the LilyGO TTGO ESP32+SIM7600 board, the DHT22, MQ-5, MQ-135, PIR and INA226 sensors inside the marine-grade enclosure, carefully routing only the sensor apertures through hydrophobic membrane vents. All I²C, UART and GPIO lines are connected to the ESP32 header per the schematics in Chapter 2.

3.2.3.2 Firmware Deployment & Configuration

Using the Arduino IDE, I uploaded the firmware that we saw in Listing ???. After reset, the board:

- Executes AT commands to power on GPS and attach GPRS via `AT+CGDCONT`, `AT+CGATT` and `AT+SAPBR`
- Reads each sensor, formats a JSON payload, and issues an HTTP POST to the Boat Pro API endpoint

I verified each step over Serial Mon, confirming network registration, GPS fix acquisition, and successful HTTP responses.

3.2.3.3 Field Testing

I conducted sea trials at 1km offshore, logging:

- GPS accuracy within 3m after 30s cold-start
- HTTP POST success >95
- Battery draw of 200mA avg in deep-sleep mode with 30s transmission intervals

3.2.3.4 Mobile App Validation

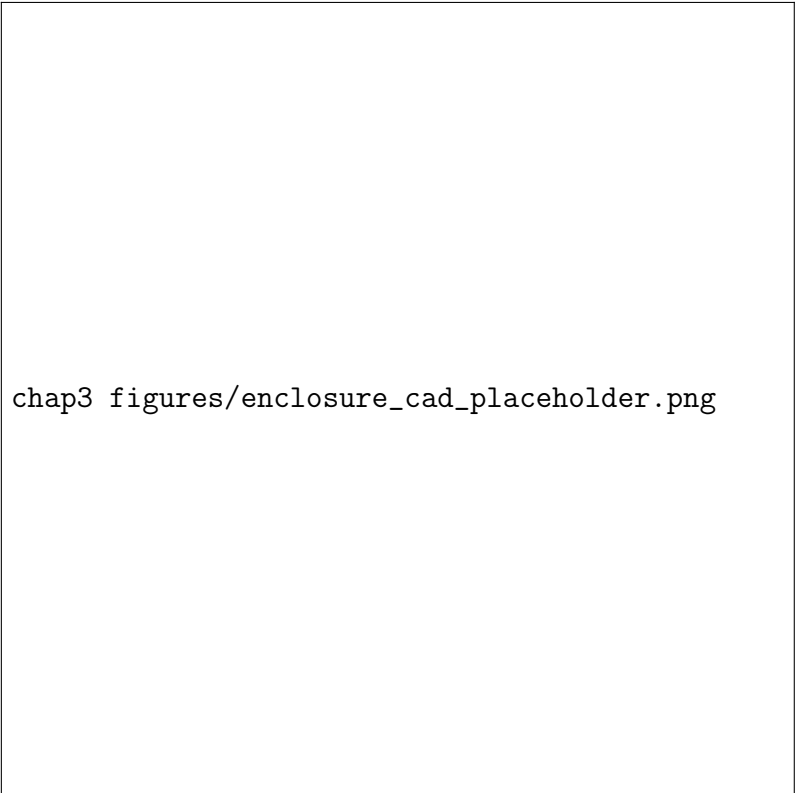
Finally, I integrated the same REST API into the Ionic/Angular Boat Pro app. Screenshots in Figure ?? show live telemetry graphs, map position, and alert banners when motion or gas thresholds are exceeded. All components worked end-to-end, demonstrating the system's readiness for real-world deployment.

3.3 Enclosure Design

Rather than designing a custom PCB, I opted to create a 3D-printed enclosure that houses the LilyGO TTGO ESP32+SIM7600 board centrally and provides dedicated mounting points and sealed apertures for each sensor module. The enclosure is designed in SolidWorks (or CAD) with the following considerations:

- **Board Mounting:** A snap-fit cradle secures the ESP32 board in the center, aligning its USB and antenna connectors with external openings.
- **Sensor Ports:** Each sensor (DHT22, MQ-5, MQ-135, PIR, INA226 shunt resistor) receives a weather-proof membrane-vented port that allows the sensing element to contact the environment while maintaining IP67 integrity.
- **Cable Routing:** Internal channels guide wiring harnesses from the board to each sensor port, minimizing strain and preventing tangling.
- **Power Access:** The 18650 battery compartment and solar input terminal are accessible through a sealed side panel for easy battery replacement and panel connection.
- **Mounting Features:** Flanges and slots on the back of the enclosure allow it to be fastened to the boat's bulkhead or railing.

Figure 3.3 shows a preliminary CAD rendering of the enclosure. I will attach a finalized SolidWorks screenshot once the design is complete.



chap3 figures/enclosure_cad_placeholder.png

Figure 3.3: 3D CAD Rendering of Enclosure (Placeholder)

3.4 Mobile Application

3.4.1 Overview of Boat Pro

Boat Pro is a social and marketplace application for boat owners, built with the Angular/Ionic framework. It provides users with:

- A chat system to communicate and share events or favorite anchorages
- A marketplace to buy, sell, and browse marine equipment
- An IoT dashboard (my scope) for real-time monitoring and control of onboard sensors

3.4.2 IoT Dashboard

The IoT page of Boat Pro is organized into a set of interactive cards. Screenshots of the live application are shown below.

3.4.2.1 Environmental Data

Displays current:

- Temperature (°C) and humidity (%)

- Motion detection status
- Gas-leak alert

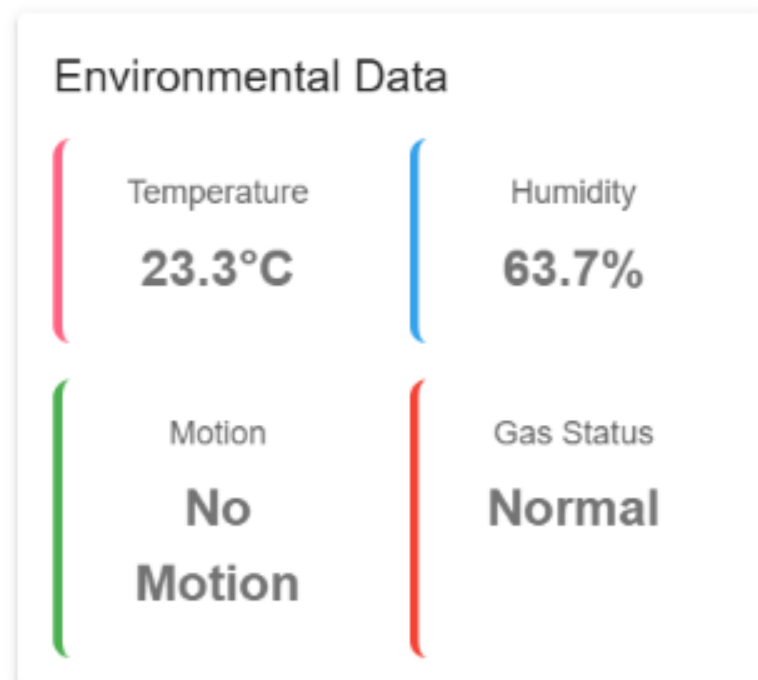


Figure 3.4: Environmental Data Card: temperature, humidity, motion, gas

3.4.2.2 System Status

Shows:

- Battery voltage (user selects 6V, 12V or 24V system)
- Boat speed (km/h) from GPS

3.4.2.3 Air Quality & CO₂

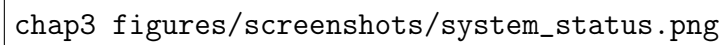
Presents:

- CO₂ concentration (ppm)
- Air quality label (Excellent, Good, Moderate, Poor, Unhealthy)

3.4.2.4 Historical Data

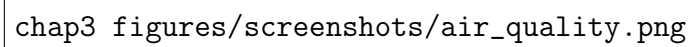
Graphs of the last 24hours for:

- Temperature (°C)
- Humidity (%)



chap3 figures/screenshots/system_status.png

Figure 3.5: System Status Card: battery voltage and speed



chap3 figures/screenshots/air_quality.png

Figure 3.6: Air Quality & CO₂ Card



chap3 figures/screenshots/history.png

Figure 3.7: Historical Data Card: 24h temperature and humidity charts

3.4.2.5 Location

Interactive map showing the boat's current GPS position.

3.4.2.6 Geofence Settings

Allows the user to:

- Enter center coordinates manually or “Reset to Boat”
- Drag radius slider from 0.1km to 10km

3.4.2.7 Geofence Events

When the boat breaches the geofence, past locations are overlaid as dots on the map, and alerts are generated every 3min until re-entry.

3.4.2.8 Notifications History

Lists all recent alerts:

- Low battery (<80%)
- Gas detection

chap3 figures/screenshots/location.png

Figure 3.8: Location Card: Real-time boat position on map

chap3 figures/screenshots/geofence_settings.png

Figure 3.9: Geofence Settings Card



chap3 figures/screenshots/geofence_events.png

Figure 3.10: Geofence Events Card: breach track overlay

- Motion (when Armed)
- Geofence breach

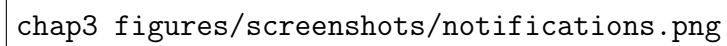
3.4.2.9 Arm/Disarm Control

Toggle for motion alerts and SMS:

- *Armed*: motion detection triggers SMS and in-app notification
- *Disarmed*: no motion alerts (user may set duration in hours/days)

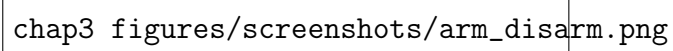
3.4.2.10 Complete IoT Dashboard Interface

Below is the full view of the Boat Pro IoT dashboard, showcasing all cards in a single screen for a holistic overview.



chap3 figures/screenshots/notifications.png

Figure 3.11: Notifications History



chap3 figures/screenshots/arm_disarm.png

Figure 3.12: Arm/Disarm Control



Figure 3.13: Complete IoT Dashboard Interface

3.5 Prototype

Our intelligent agriculture system prototype is the result of our extensive work and knowledge in developing an innovative solution. It demonstrates the successful integration of key subsystems, such as the greenhouse, sun tracker solar panel, and automatic pet watering system. These subsystems are controlled by the ESP32 microcontroller and connected through GPRS technology. The prototype embodies an advanced and automated agricultural system that enhances productivity, conserves energy, and simplifies management. It holds promise for farmers looking for a cost-effective and sustainable solution to improve their agricultural practices.

The following figures shows the real implementation of intelligent agriculture system prototype.

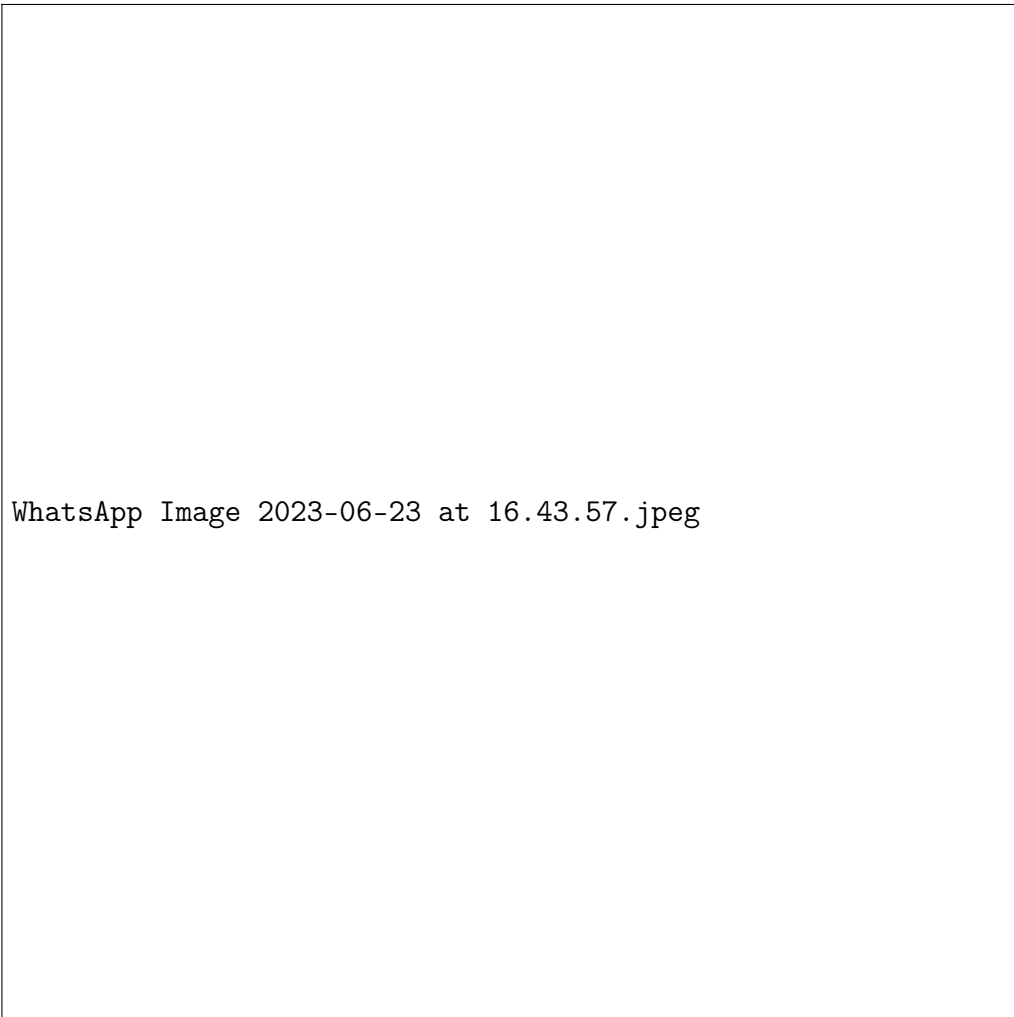
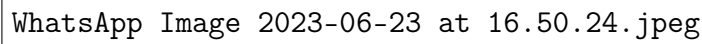


Figure 3.14: realisation and testing



WhatsApp Image 2023-06-23 at 16.50.24.jpeg

Figure 3.15: Prototype

3.6 Conclusion

In this chapter, we have provided a detailed exposition of the system, presenting an in-depth analysis of its various components. We have incorporated the circuit diagram and the schematics accompanied by a clear description of the connections. In addition, we have depicted the flowchart of our three systems, highlighting their operational processes. Furthermore, we have showcased the design of our PCB board, emphasizing its role in the system's functionality. We have explained the steps for receiving data and monitoring our system through the mobile application in a clear and understandable manner. Finally, rigorous testing and validation procedures have been undertaken to ensure the prototype's dependability and optimal performance.

General conclusion and perspectives

To complete our bachelor's degree program in Electronics, Electrotechnics, and Automation with a specialization in Embedded Systems, we were assigned a final-year project.

Our objective was to design an intelligent agricultural system that utilizes GPRS for wireless communication and the ESP32 microcontroller. This system consists of three subsystems: the greenhouse, sun tracker solar panel, and automatic pet watering.

The greenhouse subsystem maintains optimal environmental conditions for plant growth, including temperature, humidity, and soil moisture. The sun tracker solar panel optimizes solar energy utilization by adjusting its position according to the sun's movement. The automatic pet watering system ensures a continuous water supply for pets.

The integration of these subsystems has resulted in an automated agricultural system that enhances crop productivity, conserves energy, and simplifies pet care.

Our project comprised five stages. We began with a preliminary study to understand the latest technologies in the agricultural sector and determine our project's primary objective. This led to requirement analysis and specification. Once our goals were defined, we moved to the design phase.

In the second phase, we incorporated GPRS technology into our system, enabling wireless communication between the central control unit and the subsystems. This ensures real-time data transmission, efficient monitoring, and control of environmental conditions and pet watering. Next, we conducted a simulation using Fritzing and implemented our smart agriculture system. The subsystems included an irrigation system and temperature regulation system for the greenhouse, as well as pet watering and sun tracker solar panel systems. Using the ESP32 board as an interface simplified system control, as observed during practical tests.

In the fourth phase, we further enhanced our implementation by integrating the Blynk application for remote control. This allowed us to control the system elements from a distance or via the internet using the ESP32 and GPRS module.

Finally, we completed the implementation phase, gaining familiarity with the

General conclusion and perspectives

coding aspects of the intelligent agricultural system for remote control and monitoring.

Then we are pleased with the successful accomplishment of our final year project, which has the potential to advance the agricultural sector by providing cost-effective and sustainable improvements to working conditions. Our perspective for this project is :

- Integration of photovoltaic panels: We plan to incorporate photovoltaic panels into the system to significantly reduce the overall electrical consumption by harnessing solar energy, to ensure a more sustainable and environmentally friendly operation.
- Implementation of LoRa technology: Instead of relying on traditional GPRS technology, we have chosen to utilize LoRa for long-range communication in our IoT applications. LoRa offers an energy-efficient solution, allowing connectivity over vast areas with minimal power consumption. This choice ensures optimized power management and enables seamless communication in our system.
- Utilization of STM32 microcontroller: The selection of the STM32 microcontroller is driven by its renowned low-power operation capabilities. By leveraging this microcontroller, we can achieve efficient energy management and optimize the overall power consumption of our system.
- Development of a customized sensor node: In order to optimize system performance and meet our specific requirements, we will design and integrate a bespoke sensor node. This tailor-made solution will enable precise data collection and monitoring, while also reducing the need for excessive wiring and cables.

Webography

- [1] “Dht22 temperature and humidity sensor module image.” Manufacturer’s product photo.
- [2] “Dht22 pinout diagram.” Adapted from multiple online sources.
- [3] “Hc-sr501 pir motion sensor module image.” Manufacturer’s product photo.
- [4] “Hc-sr501 pir sensor pinout diagram.” Adapted from multiple sources.
- [5] “Ina226 current and power monitor module image.” TinyTronics image.
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- [7] “Mq-5 combustible gas sensor module image.” Accessed 2025-05-13.
- [8] “Mq-135 air quality sensor image.” Accessed 2025-05-13.
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